INTRODUCTION

In today's data-driven world, organizations rely heavily on data analysis to make informed business decisions and improve overall performance. The ability to transform raw data into meaningful insights plays a crucial role in understanding customer behavior, tracking sales performance, and identifying growth opportunities. Business Intelligence (BI) tools have become essential for analyzing large datasets and presenting them in a visually understandable format.

This project focuses on the development of a Blinkit Sales Dashboard using SQL and Power BI. Blinkit, a quick-commerce platform, generates a large amount of sales and customer data on a daily basis. Analyzing this data helps in identifying patterns such as top-selling products, sales trends, and customer preferences. The main objective of this project is to convert raw sales data into an interactive dashboard that provides clear insights into business performance.

The project involves multiple stages, starting from data extraction and processing using SQL, followed by data cleaning and transformation using Power Query. After preparing the dataset, Power BI is used to create visualizations and dashboards that represent key metrics such as total sales, profit, quantity sold, and average order value. These visual elements help in presenting complex data in a simplified and user-friendly manner.

In addition, the project utilizes DAX (Data Analysis Expressions) for calculating key performance indicators (KPIs) and enhancing the analytical capabilities of the dashboard. The integration of SQL and Power BI ensures efficient data handling, accurate analysis, and real-time insights, which are essential for effective decision-making in modern businesses.

Overall, this project demonstrates how data analysis and visualization tools can be used to transform raw data into actionable insights. It highlights the importance of business intelligence in improving operational efficiency, understanding market trends, and supporting strategic planning in a competitive business environment.

PROJECT DESCRIPTION

During my internship at Technogrowth Software Solutions Pvt. Ltd., I worked on data analysis and visualization using Power BI. The project focused on developing an interactive Blinkit Sales Dashboard to transform raw business data into meaningful insights for decision-making.

Blinkit is an online grocery delivery platform that generates large volumes of transactional data. Analyzing this data is important to understand sales performance, customer behavior, and product trends. The main objective of this project was to design a dashboard that provides a clear and structured view of business performance using key metrics and visualizations.

The project involved multiple stages including data loading, data cleaning, data transformation, KPI calculation, and dashboard development. Data cleaning and preprocessing were performed using Power Query in Power BI, where missing values were handled, data types were corrected, and unnecessary data was removed to ensure accuracy and consistency.

After preparing the dataset, key performance indicators (KPIs) such as Total Sales, Average Sales, Number of Items Sold, and Average Rating were calculated using DAX formulas. These KPIs provide a quick overview of business performance and help in tracking important metrics. Finally, an interactive dashboard was created using Power BI. The dashboard includes various

visualizations such as bar charts, pie charts, and line charts to represent data effectively. It also contains filters and slicers that allow users to interact with the data and analyze specific segments such as outlet size, location, and product category.

Key Features :

- **KPI Cards** : Displays Total Sales, Average Sales, Number of Items Sold, and Average Rating
- **Sales by Outlet Location**: Shows performance across different locations
- **Sales by Outlet Size**: Compares small, medium, and large outlets
- **Sales by Item Type**: Identifies top-performing product categories
- **Fat Content Analysis**: Compares low-fat and regular products
- **Trend Analysis**: Shows outlet establishment trends over time
- **Interactive Dashboard**: Includes filters and slicers for dynamic analysis

Tech Stack Used :

- Power BI – Dashboard creation and visualization
- Power Query – Data cleaning and transformation
- DAX – KPI calculation and measures

SYSTEM ANALYSIS

Existing System :

The existing system relies on raw data stored in datasets or spreadsheets, which is difficult to analyze directly.

- Data is in tabular format and not easy to understand.
- No proper visualization of sales performance
- Manual analysis is time-consuming
- Difficult to identify trends and patterns
- Limited support for decision-making

Proposed System :

The proposed system uses Power BI to create an interactive Blinkit Sales Dashboard for better analysis and visualization.

- Interactive dashboard for easy data understanding
- Visual representation using charts and graphs
- KPI-based performance tracking (Sales, Items, Ratings)
- Faster and accurate decision-making
- User-friendly interface with filters and slicers

System Objectives :

- To analyze sales data effectively
- To create an interactive dashboard
- To track KPIs like sales and ratings
- To support data-driven decision-making

System Workflow :

The system follows a structured process to convert raw data into meaningful insights:

1. Data Loading: Import dataset into Power BI
2. Data Cleaning: Use Power Query to handle missing values and correct data types
3. Data Transformation: Prepare structured data for analysis
4. KPI Calculation: Use DAX to calculate metrics like Total Sales and Average Sales
5. Dashboard Creation: Design charts and visuals in Power BI
6. Insight Generation: Analyze data and identify trends

SOFTWARE REQUIREMENTS SPECIFICATION

Software Requirements:

The following software tools were used for developing the Blinkit Sales Dashboard:

- Power BI Desktop: Used for creating interactive dashboards and visualizations
- Power Query: Used for data cleaning and transformation
- DAX (Data Analysis Expressions): Used for calculating KPIs and measures

Hardware Requirements:

The following hardware configuration was used:

- System: Laptop / Personal Computer
- Processor: Minimum Intel i5 or equivalent
- RAM: 8 GB or above
- Storage: Minimum 256 GB SSD
- Operating System: Windows 10 or above

TECHNOLOGIES

The Blinkit Sales Dashboard project was developed using modern data visualization and data transformation tools. These technologies helped in converting raw data into meaningful insights and interactive reports. The main tools used in this project were Power BI, Power Query, and DAX.

Power BI :

Power BI is a business intelligence and data visualization tool used to create interactive dashboards and reports. It helps organizations analyze data and make informed decisions.

Features of PowerBI

- Interactive dashboards with multiple visualizations
- Easy drag-and-drop interface
- Real-time data analysis capabilities
- Integration with various data sources
- Support for filters and slicers for dynamic analysis

Usage in Project

- Developed Blinkit Sales Dashboard
- Created visualizations such as bar charts, pie charts, and line charts
- Designed KPI cards to display key metrics
- Added filters and slicers for better user interaction
- Analyzed sales performance across categories and locations

Power Query : Power Query is a data transformation and preparation tool available in Power BI. It is used to clean and structure raw data before analysis. **Features of Power Query**

- Data cleaning and transformation
- Handling missing and null values
- Changing data types

- Removing duplicates and unnecessary data
- Filtering and sorting data

Usage in Project

- Cleaned raw dataset before visualization
- Removed null and inconsistent values
- Corrected data types for accurate analysis
- Prepared structured data for dashboard creation

DAX (Data Analysis Expressions) :

DAX is a formula language used in Power BI to create calculated columns and measures. It is used for performing advanced data analysis. **Features of DAX**

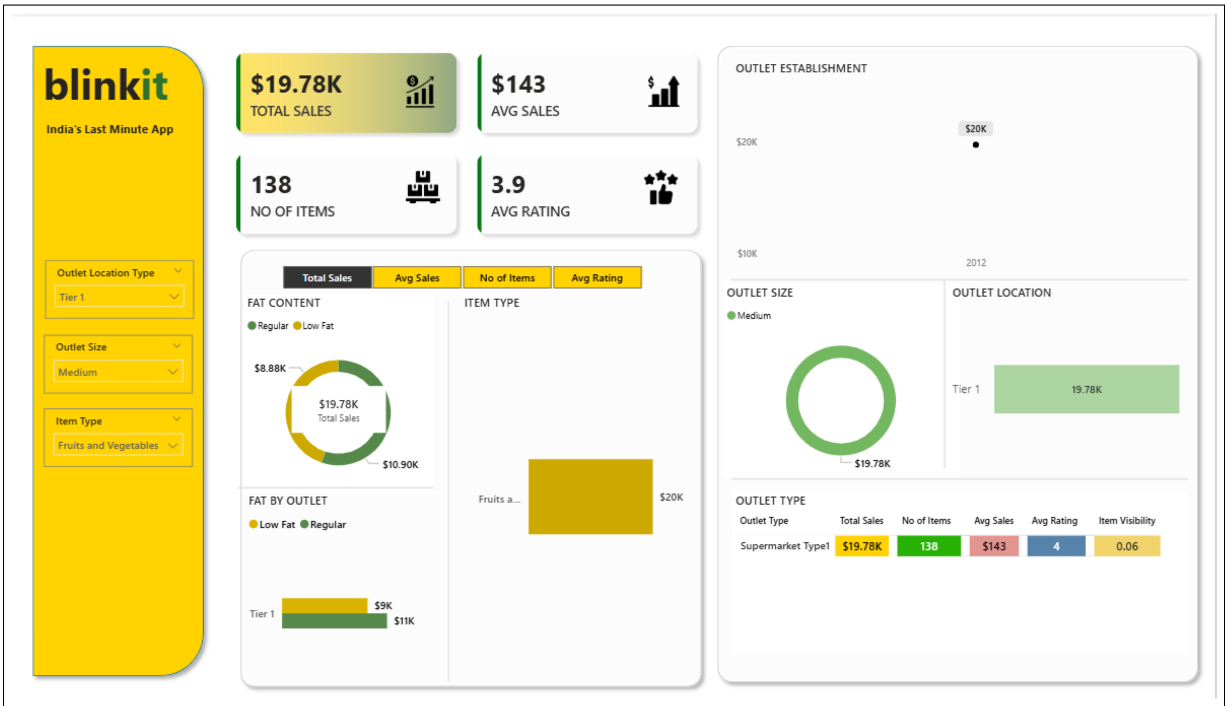
- Supports functions like SUM, AVERAGE, COUNT
- Used for creating KPIs and measures
- Enables dynamic calculations
- Helps in data aggregation and analysis

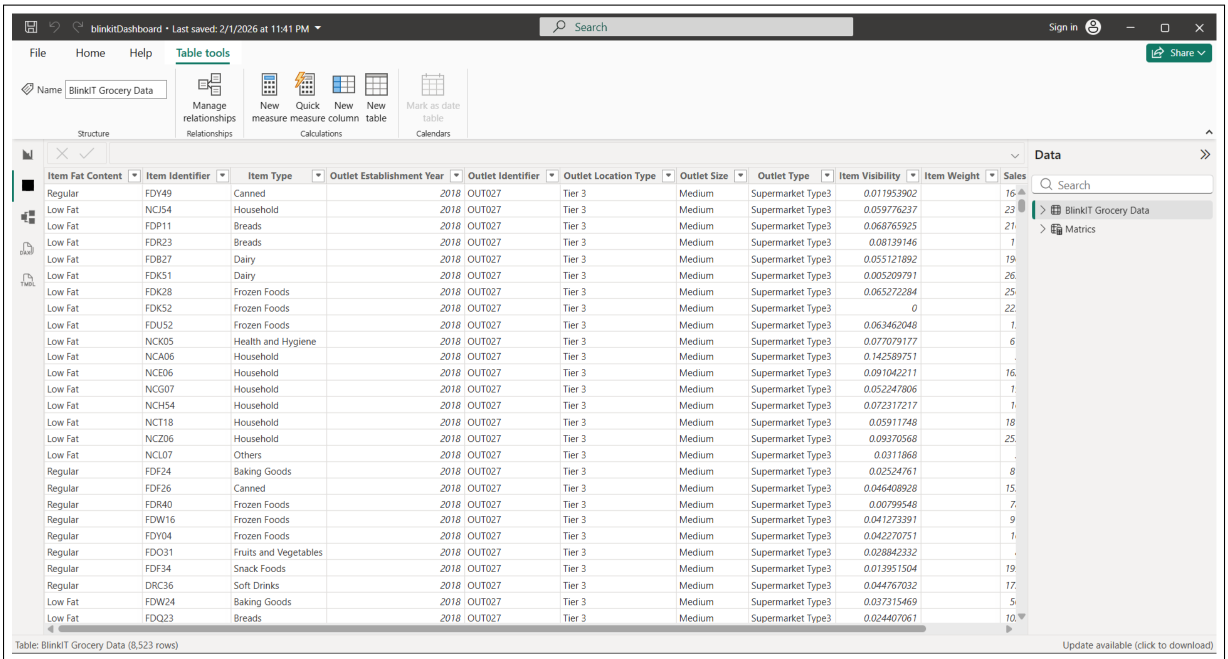
Usage in Project

- Calculated Total Sales
- Calculated Average Sales
- Calculated Number of Items Sold
- Calculated Average Rating
- Created KPI measures for dashboard

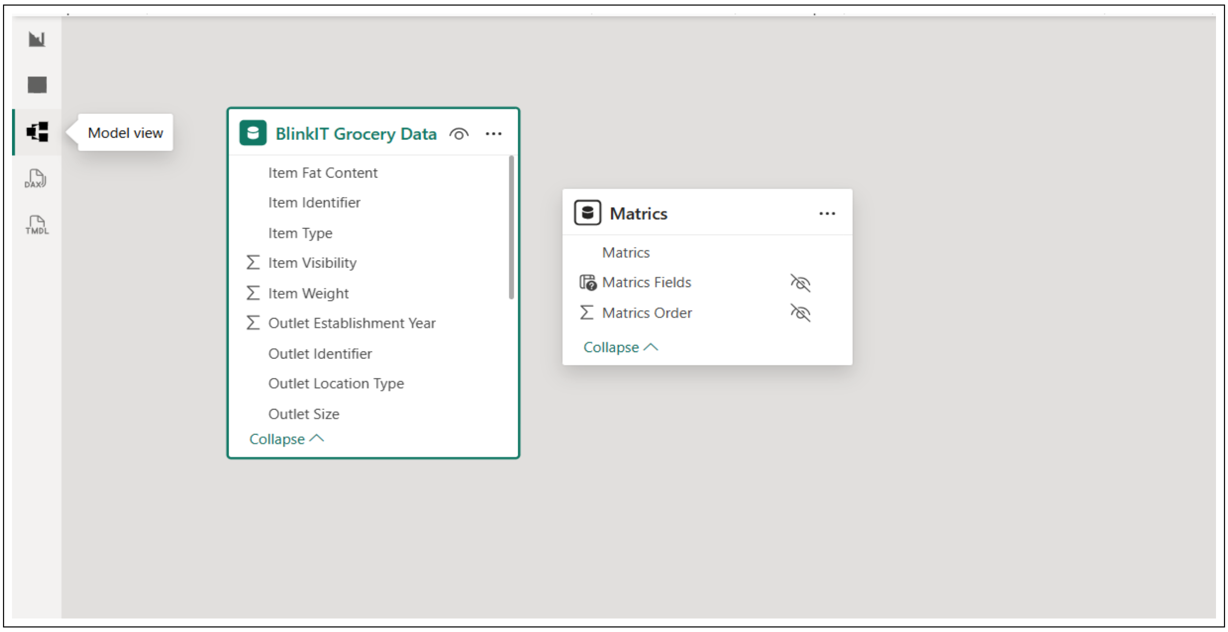
SCREENSHOTS

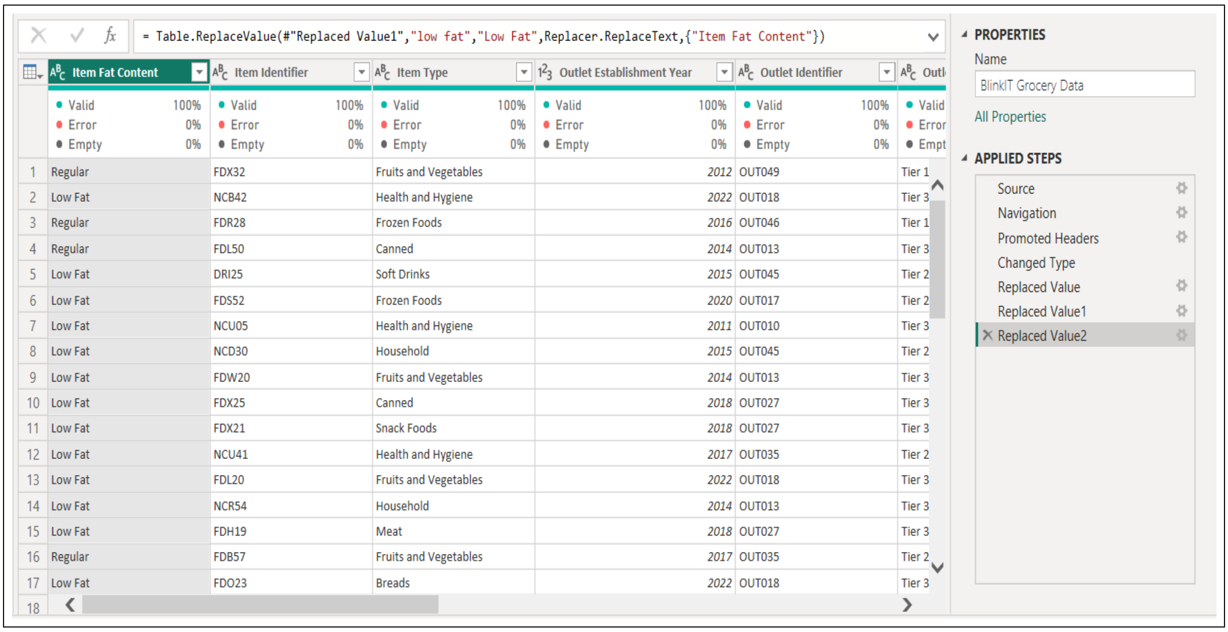
Outlet Type	Total Sales	No of Items	Avg Sales	Avg Rating	Item Visibility
Grocery Store	$151.94K	1083	$140	4	0.10
Supermarket Type1	$787.55K	5577	$141	4	0.06
Supermarket Type2	$131.48K	928	$142	4	0.06
Supermarket Type3	$130.71K	935	$140	4	0.06





Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outlet Location Type	Outlet Size	Outlet Type	Item Visibility	Item Weight	Sales
Regular	FDY49	Canned	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.011953902		16
Low Fat	NCJ54	Household	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.059776237		23
Low Fat	FDP11	Breads	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.068765925		21
Low Fat	FDR23	Breads	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.08139146		1
Low Fat	FDB27	Dairy	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.055121892		19
Low Fat	FDK51	Dairy	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.005209791		26.
Low Fat	FDK28	Frozen Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.065272284		25
Low Fat	FDK52	Frozen Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0		22.
Low Fat	FDU52	Frozen Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.063462048		1.
Low Fat	NCK05	Health and Hygiene	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.077079177		6
Low Fat	NCA06	Household	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.142589751		.
Low Fat	NCE06	Household	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.091042211		16.
Low Fat	NCG07	Household	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.052247806		1.
Low Fat	NCH54	Household	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.072317217		1
Low Fat	NCT18	Household	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.05911748		18
Low Fat	NCZ06	Household	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.09370568		25.
Low Fat	NCL07	Others	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.0311868		.
Regular	FDF24	Baking Goods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.02524761		8
Regular	FDF26	Canned	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.046408928		15.
Regular	FDR40	Frozen Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.00799548		7.
Regular	FDW16	Frozen Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.041273391		9
Regular	FDY04	Frozen Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.042270751		1
Regular	FDO31	Fruits and Vegetables	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.028842332		.
Regular	FDF34	Snack Foods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.013951504		19.
Regular	DRC36	Soft Drinks	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.044767032		17.
Low Fat	FDW24	Baking Goods	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.037315469		5
Low Fat	FDQ23	Breads	2018	OUT027	Tier 3	Medium	Supermarket Type3	0.024407061		10.





	Item Fat Content	Item Identifier	Item Type	Outlet Establishment Year	Outlet Identifier	Outl
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empt
1	Regular	FDX32	Fruits and Vegetables	2012	OUT049	Tier 1
2	Low Fat	NCB42	Health and Hygiene	2022	OUT018	Tier 3
3	Regular	FDR28	Frozen Foods	2016	OUT046	Tier 1
4	Regular	FDL50	Canned	2014	OUT013	Tier 3
5	Low Fat	DRI25	Soft Drinks	2015	OUT045	Tier 2
6	Low Fat	FDS52	Frozen Foods	2020	OUT017	Tier 2
7	Low Fat	NCU05	Health and Hygiene	2011	OUT010	Tier 3
8	Low Fat	NCD30	Household	2015	OUT045	Tier 2
9	Low Fat	FDW20	Fruits and Vegetables	2014	OUT013	Tier 3
10	Low Fat	FDX25	Canned	2018	OUT027	Tier 3
11	Low Fat	FDX21	Snack Foods	2018	OUT027	Tier 3
12	Low Fat	NCU41	Health and Hygiene	2017	OUT035	Tier 2
13	Low Fat	FDL20	Fruits and Vegetables	2022	OUT018	Tier 3
14	Low Fat	NCR54	Household	2014	OUT013	Tier 3
15	Low Fat	FDH19	Meat	2018	OUT027	Tier 3
16	Regular	FDB57	Fruits and Vegetables	2017	OUT035	Tier 2
17	Low Fat	FDO23	Breads	2022	OUT018	Tier 3
18						

CONCLUSION

The **Blinkit Sales Dashboard** project successfully demonstrates the importance of data analysis and visualization in understanding business performance. By using SQL and Power BI, raw sales data was transformed into meaningful insights that help in identifying trends, patterns, and key performance indicators. The dashboard provides a clear and interactive way to monitor important metrics such as total sales, profit, quantity sold, and average order value.

Throughout the project, various steps such as data extraction, data cleaning, and data transformation were carried out to ensure data accuracy and reliability. **Power BI** played a crucial role in creating visual representations of data, making it easier to interpret complex information. The use of **Power Query** and **DAX** further enhanced the analytical capabilities by enabling efficient data processing and calculation of KPIs.

This project also helped in understanding how businesses can use dashboards to make informed decisions. By analyzing sales data across different categories, locations, and time periods, organizations can identify high-performing areas as well as opportunities for improvement. The interactive nature of the dashboard allows users to explore data dynamically and gain deeper insights.

In addition to technical skills, the project improved analytical thinking, problem-solving abilities, and attention to detail. Working with real-world data provided practical exposure to data analysis processes and tools used in the industry. It also strengthened the understanding of how data-driven approaches are applied in business environments.

Overall, the project highlights the significance of business intelligence tools in modern organizations. It demonstrates how effective data analysis and visualization can support better decision-making, improve efficiency, and contribute to business growth. This experience serves as a strong foundation for future work in data analysis and related fields.

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